

Task 1: NullPointerException

Scenario:

You are building a student profile system. Sometimes, student data may not be available.

Task:

- Create a String `studentName = null;`
- Try to print the length of the name using `.length()`
- Handle the exception using try-catch

Task 2: ArrayIndexOutOfBoundsException

Scenario:

You are storing marks of 3 subjects, but user tries to access invalid index.

Task:

- Create an array: `int marks[] = {85, 90, 78};`
- Try to access `marks[5]`
- Handle the exception

Task 3: ArithmeticException

Scenario:

You are building a calculator where user enters two numbers.

Task:

- Take two integers:

int a = 10;

int b = 0;

- Perform division: a / b
- Handle the exception

```
Day-8 > J exe.java > exe > main(String[] args)
1  class exe{
    Run main | Debug main
2  public static void main(String[] args) {
3      //ArithmeticException
4      int a = 77;
5      try {
6          int b = a/0;
7          System.out.println(b);
8      }
9      catch (ArithmeticException e) {
10         System.out.println(e);
11     }
12     //Null pointer
13     String s = null;
14     try {
15         System.out.println(s.length());
16     } catch (NullPointerException e) {
17         System.out.println(e);
18     }
19     //index out of bounds
20     int arr[]={1,2,3,4,5};
21
22     try {
23         arr[9]=7;
24     }
25     catch (ArrayIndexOutOfBoundsException e) {
26         System.out.println(e);
27     }
28     String str[]={};
29     try {
30         System.out.println(str[7]);
31     } catch (ArrayIndexOutOfBoundsException e) {
32         System.out.println(e);
33     }
34
35 }
36 }
```

```
PS D:\Java_161\Day-8> javac exe.java
PS D:\Java_161\Day-8> java exe
java.lang.ArithmeticException: / by zero
java.lang.NullPointerException: Cannot invoke "String.length()" because "<local2>" is null
java.lang.ArrayIndexOutOfBoundsException: Index 9 out of bounds for length 5
java.lang.ArrayIndexOutOfBoundsException: Index 7 out of bounds for length 0
PS D:\Java_161\Day-8> █
```